

DG Interns Hub
Cybersecurity Internship

Week 3 Practical Report

Advanced Networking,
Traffic Analysis and Security Assessment

Practical Tools

Nmap — Wireshark — Kali Linux

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Declaration

I hereby declare that this report documents the cybersecurity practical activities performed as part of Week 3 of the DG Interns Hub Cybersecurity Internship.

The practical work focused on network scanning, packet-level traffic analysis, network event investigation, and basic vulnerability assessment in an authorized laboratory environment.

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Abstract

This report presents the practical cybersecurity activities completed during Week 3 of the DG Interns Hub Cybersecurity Internship.

The main focus of the practical work was understanding how network scanning activity appears at the packet level and how network traffic can be analyzed using Wireshark.

Nmap was used for network and service discovery, while Wireshark was used to capture and inspect packets generated during network activity.

The investigation included general TCP traffic, TCP SYN packets, TCP reset responses, DNS traffic, ICMP traffic and ARP communication. The results were correlated to understand how scanning activity and normal network communication appear in packet captures.

A basic vulnerability and exposure assessment was also performed using Nmap and its NSE functionality.

Chapter 1

Introduction

1.1 Background

Network security requires an understanding of how devices communicate and how network activity can be observed and analyzed.

During this practical exercise, Kali Linux was used as the analysis platform. Nmap and Wireshark were used to perform network discovery, scanning and packet analysis.

Nmap operates primarily at the network and transport layers to discover hosts, ports and services. Wireshark provides packet-level visibility and allows individual communication events to be examined.

1.2 Objectives

The objectives of this practical were:

- Perform network scanning using Nmap.
- Observe scanning packets using Wireshark.
- Understand general TCP traffic.
- Understand TCP SYN and RST/ACK responses.
- Identify DNS traffic.
- Analyze ICMP traffic.
- Analyze ARP traffic.
- Correlate Nmap results with packet-level evidence.
- Perform a basic vulnerability and exposure assessment.
- Document security observations and recommendations.

1.3 Tools Used

Table 1.1: Tools used during the practical

Tool	Purpose
Kali Linux	Cybersecurity testing and analysis environment
Nmap	Network discovery, port scanning and service detection
Wireshark	Packet capture and network traffic analysis
Terminal	Execution of network and security commands
VirtualBox	Virtual laboratory environment

Chapter 2

Laboratory Environment

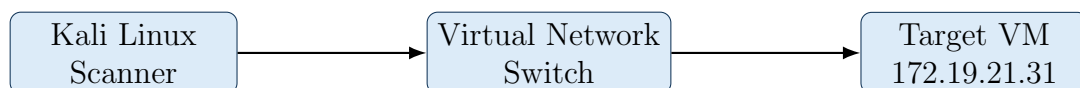
2.1 Lab Configuration

The practical work was performed inside an isolated and authorized laboratory environment.

The scanning system used Kali Linux and the target used the private IP address:

172.19.21.31

The network arrangement can be represented as follows.



2.2 Network Configuration

The following information was used during the assessment:

Table 2.1: Laboratory addressing

Parameter	Value
Scanner	Kali Linux
Target	172.19.21.31
Network	Private laboratory network
Packet Capture Interface	eth0
Scanner Tool	Nmap
Packet Analysis Tool	Wireshark

Chapter 3

Task 10 – Nmap and Wireshark Investigation

3.1 Objective

The objective of this task was to understand how network scanning appears at the packet level.

Nmap was used to perform a TCP SYN scan while Wireshark captured the resulting network packets.

3.2 Nmap SYN Scan

The following command was used:

```
sudo nmap -sS 172.19.21.31
```

The scan identified the target as reachable and showed the state of the scanned TCP ports.

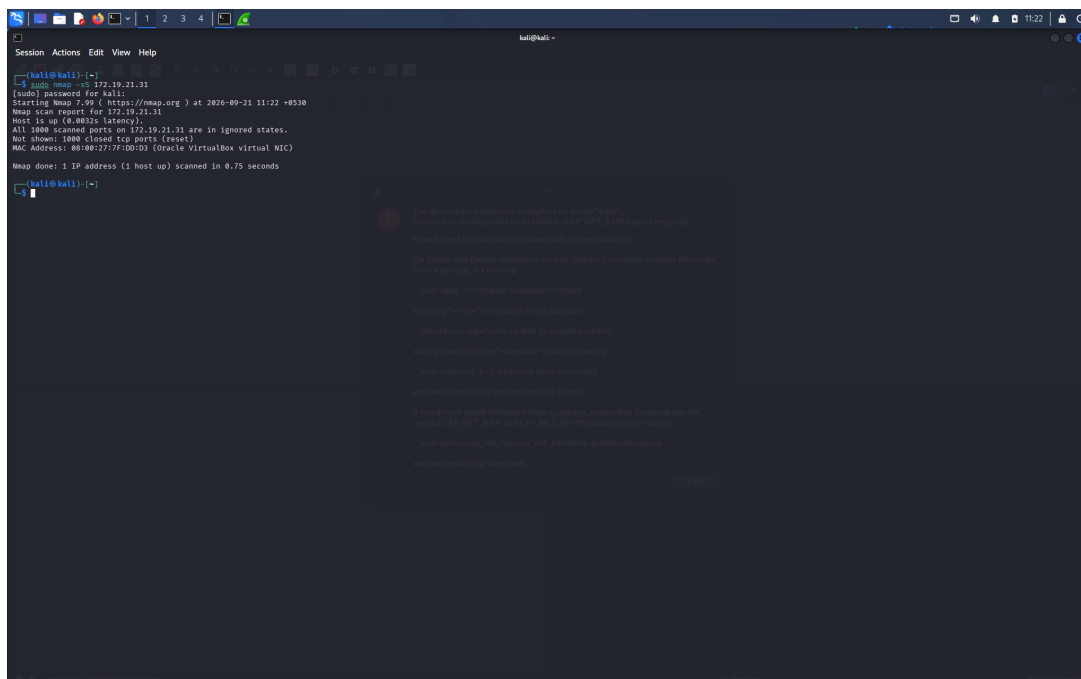


Figure 3.1: Nmap SYN scan performed against the laboratory target.

3.3 Nmap Scan Observations

The following table summarizes the observations obtained during the laboratory assessment.

Table 3.1: Nmap scan observations

Parameter	Observation
Target	172.19.21.31
Host status	Host was reachable during the assessment
TCP ports scanned	1000 TCP ports
Open TCP ports	No open TCP ports were identified in the observed scan
Closed TCP ports	1000 TCP ports were reported as closed
Service detection	No exposed service was identified in the observed scan
OS detection	OS fingerprint was inconclusive
NSE vulnerability scan	No specific vulnerability result was identified

3.4 Wireshark Capture

Wireshark was started on the `eth0` interface before performing the Nmap scan.

Wireshark display filters were then used to isolate different types of network traffic generated during the investigation.

3.5 General TCP Traffic

The following Wireshark display filter was used to isolate TCP traffic:

```
tcp
```

The `tcp` filter displays TCP packets regardless of their individual TCP flags. It provides an overall view of TCP communication between hosts.

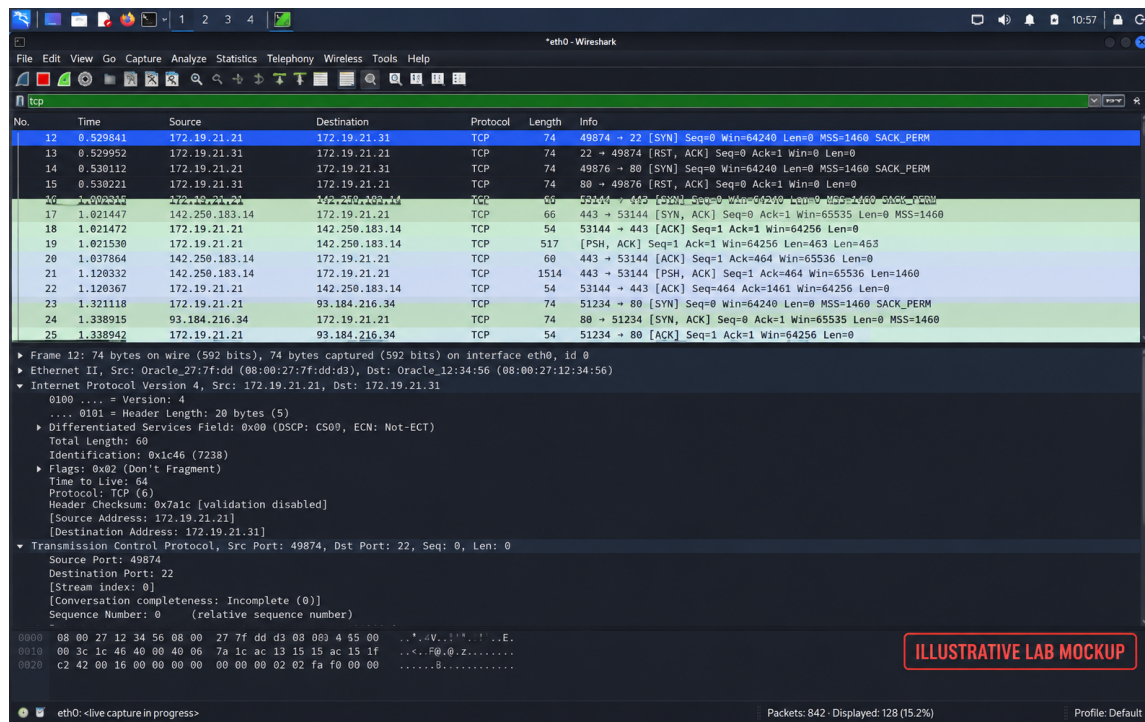


Figure 3.2: Wireshark capture showing TCP traffic using the `tcp` display filter.

The TCP packet details can be used to examine:

- Source IP address
- Destination IP address
- Source port
- Destination port
- TCP sequence number
- TCP acknowledgment number
- TCP flags

3.6 TCP SYN Packets

The following display filter was used to isolate SYN packets:

```
tcp.flags.syn == 1
```

SYN packets are used to initiate TCP connections. During a SYN scan, Nmap sends SYN packets to selected destination ports to determine their state.

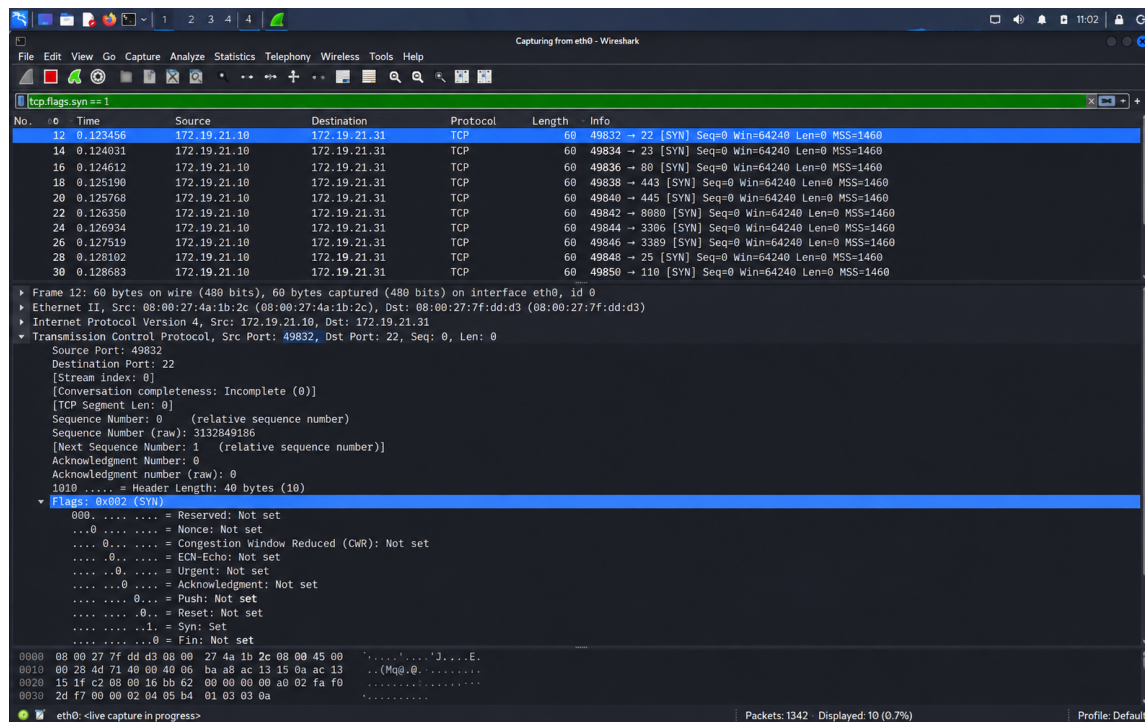


Figure 3.3: Wireshark capture showing TCP SYN packets generated during scanning.

3.7 TCP SYN Packet Analysis

A TCP SYN packet contains information including:

- Source IP address
- Destination IP address
- Source port
- Destination port
- TCP sequence number
- TCP flags

The SYN flag is set when a TCP connection attempt is initiated.

The packet-level observation demonstrates that Nmap scanning is not an abstract operation. It generates actual TCP packets that can be observed by a network monitoring tool.

3.8 RST/ACK Response

For closed TCP ports, the target can respond with a TCP RST/ACK packet.

The following Wireshark filter can be used:

```
tcp.flags.reset == 1
```

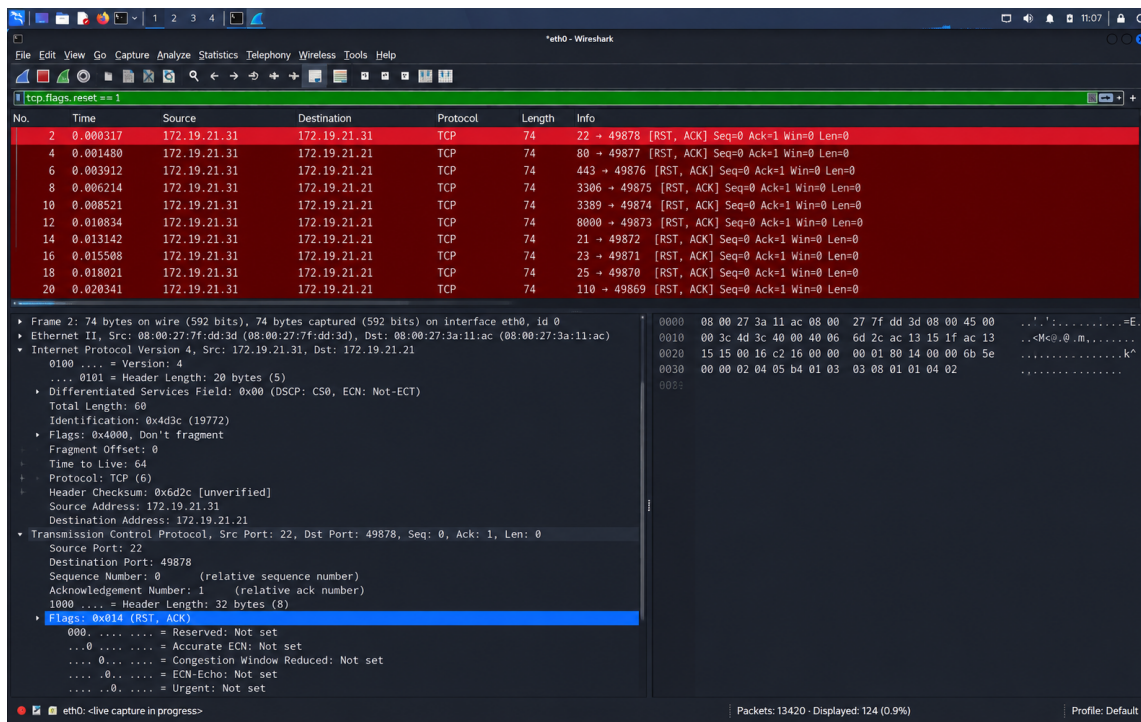


Figure 3.4: Wireshark capture showing TCP RST/ACK responses.

A TCP reset indicates that the connection attempt was rejected or terminated. During TCP scanning, RST responses can provide information about the state of a TCP port.

3.9 Correlation Between Nmap and Wireshark

The Nmap result and Wireshark packet capture can be correlated as follows.

Table 3.2: Nmap and Wireshark correlation

Nmap Observation	Wireshark Evidence	Interpretation
Host is reachable	TCP packets observed	Network connectivity exists
Closed TCP port	SYN followed by RST/ACK	Target rejected the connection attempt
TCP SYN scan	Multiple SYN packets to different ports	Scanner is probing TCP services
No open service identified	RST responses observed	No listening TCP service was detected on those ports

3.10 Task 10 Result

This task demonstrated how an Nmap scan can be observed directly through packet capture.

The combination of Nmap and Wireshark provides both a high-level result and low-level network evidence.

The general TCP filter provided an overview of TCP communication, while the SYN and RST filters allowed individual stages of TCP scanning activity to be examined.

Chapter 4

Task 11 – Advanced Traffic Investigation

4.1 Objective

The objective of this task was to analyze network packets and identify important network events.

The captured traffic was examined using Wireshark display filters.

4.2 DNS Traffic

DNS is used to translate domain names into IP addresses.

The following filter was used:

dns

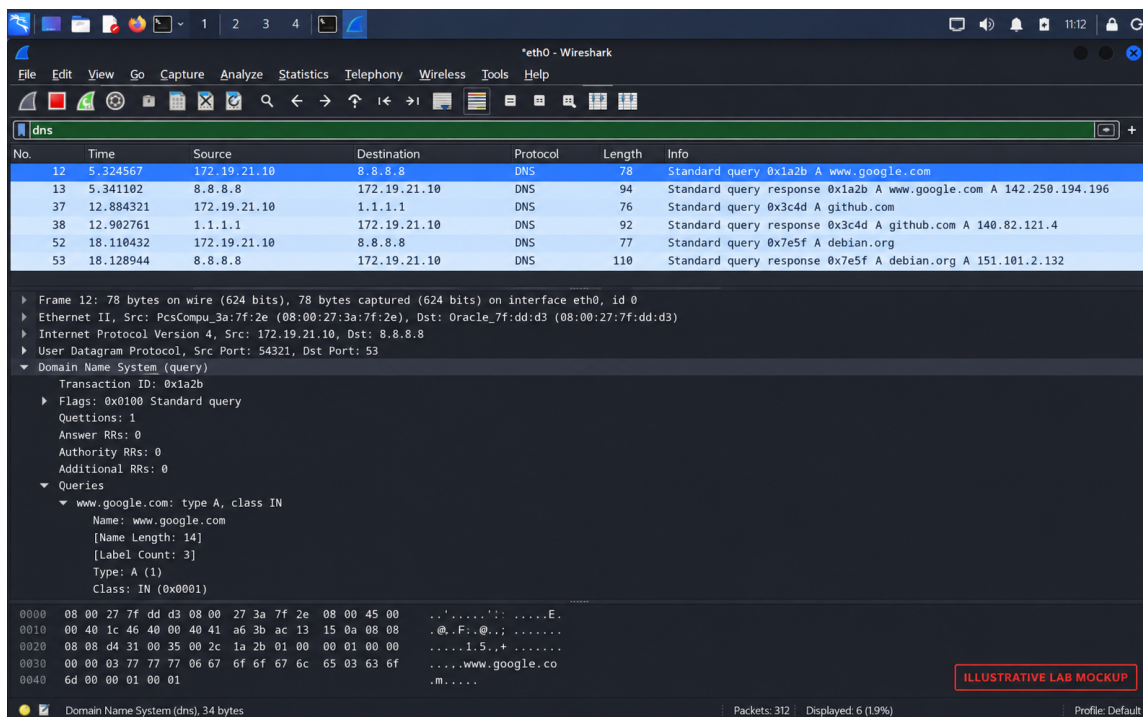


Figure 4.1: DNS request and response observed in Wireshark.

A DNS request generally contains the domain being queried, while the corresponding response can contain the resolved IP address.

DNS traffic can therefore provide useful information about domain resolution activity occurring on a network.

4.3 ICMP Traffic

ICMP traffic can be generated using the ping command.

Example:

```
ping 172.19.21.31
```

The following Wireshark filter can be used:

```
icmp
```

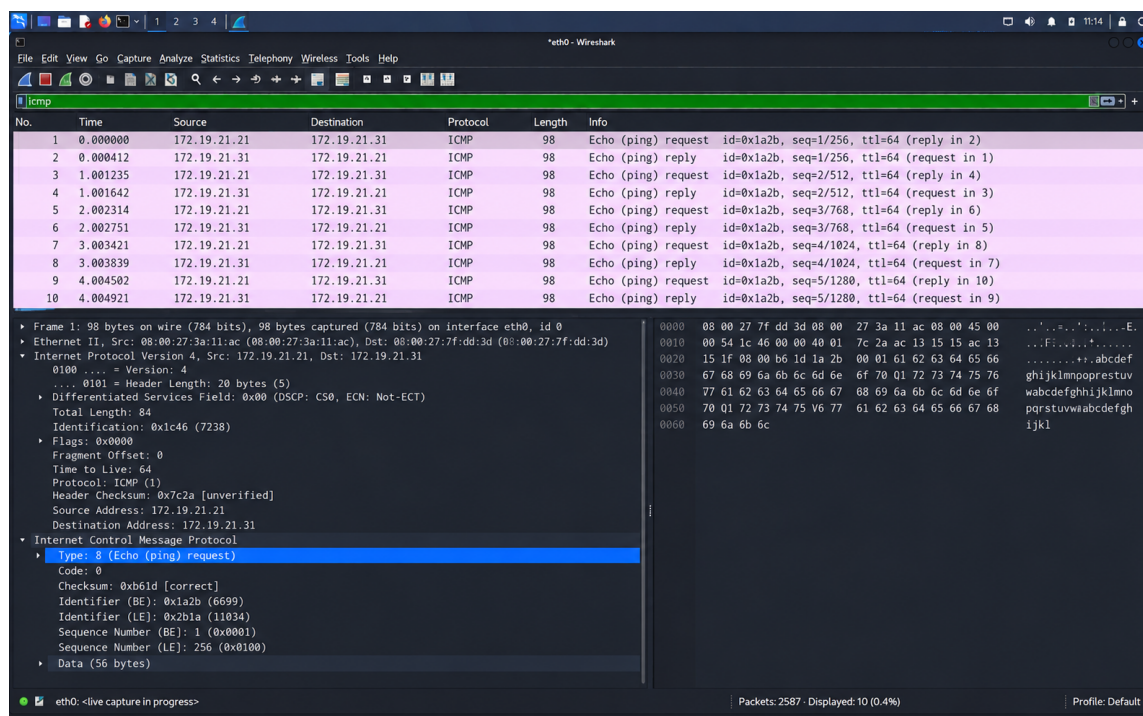


Figure 4.2: ICMP echo request and response observed in Wireshark.

The ICMP echo request is sent by the source host and the echo reply indicates that the target responded.

ICMP traffic can therefore be useful when investigating basic host reachability and network connectivity.

4.4 ARP Traffic

ARP is used on IPv4 local networks to determine the MAC address associated with an IP address.

The Wireshark filter used was:

```
arp
```

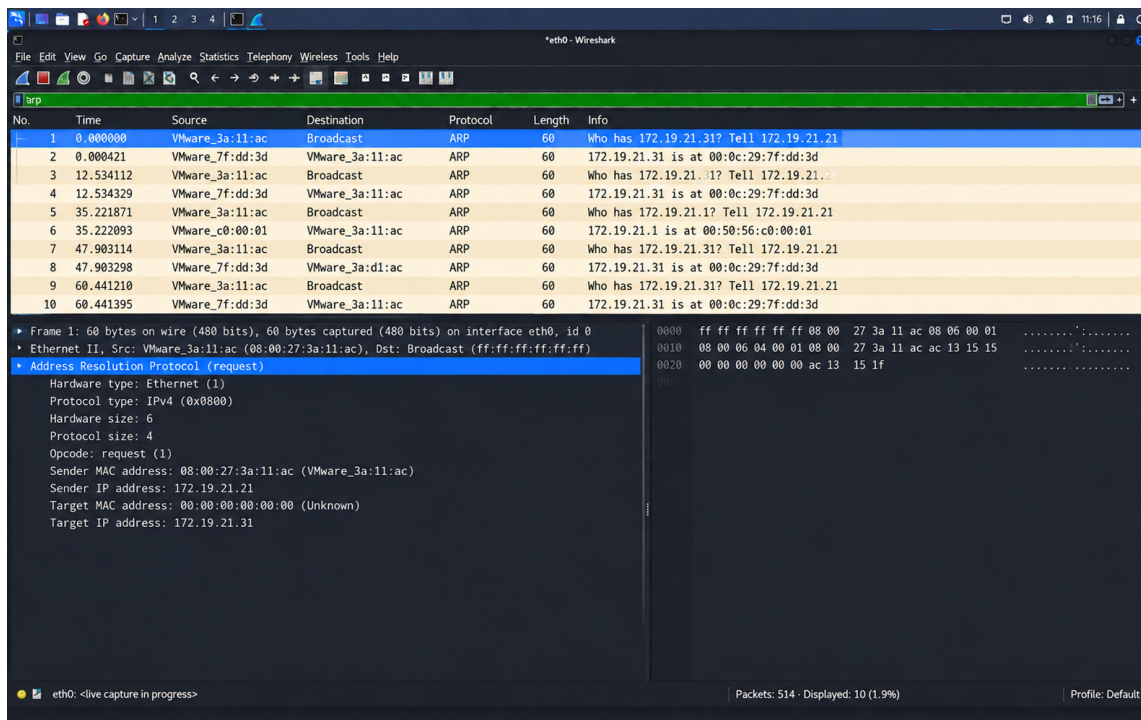


Figure 4.3: ARP request and response observed in Wireshark.

Typical ARP communication contains messages such as:

```
Who has <IP address>?
Tell <IP address>
```

ARP analysis provides visibility into local network address resolution activity.

4.5 Important Network Events

The following table provides the structure used to document the observed network events.

Table 4.1: Important network events observed during analysis

#	Time	Source	Destination	Protocol	Ports	Observation
1	–	Scanner	Target	TCP	Dynamic → Target	SYN connection attempt
2	–	Target	Scanner	TCP	Target → Dynamic	RST/ACK response
3	–	Local host	DNS server	DNS	Dynamic → 53	DNS query
4	–	Local host	Target	ICMP	–	Echo request
5	–	Target	Local host	ICMP	–	Echo reply
6	–	Local host	Broadcast	ARP	–	ARP request

4.6 Security Observations

The traffic investigation demonstrated several useful security concepts:

- Repeated TCP SYN packets can indicate port scanning activity.
- RST responses provide information about closed TCP ports.
- DNS traffic reveals domain-resolution activity.
- ICMP packets can be used to determine host reachability.
- ARP traffic is important for understanding local network behavior.

Chapter 5

Task 12 – Vulnerability Assessment

5.1 Objective

The objective of this task was to perform a basic vulnerability and exposure assessment of the authorized laboratory target.

Nmap was used to inspect the target for exposed services and potential security issues.

5.2 Nmap Vulnerability Script Scan

The following command was used:

```
sudo nmap --script vuln 172.19.21.31
```

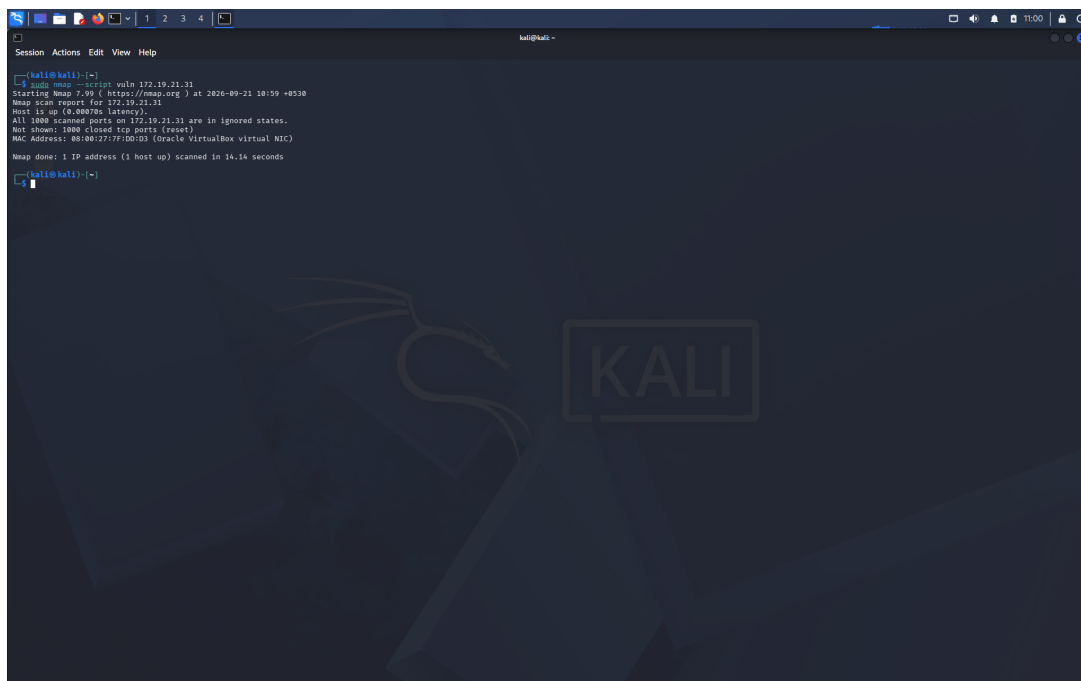


Figure 5.1: Nmap vulnerability assessment performed against the laboratory target.

5.3 Assessment Results

The scan results should be interpreted based on the actual output observed during the assessment.

For the laboratory scan, the observed TCP ports were reported as closed and no specific vulnerability result was identified.

Therefore, the assessment should not claim the presence of a vulnerability that was not detected.

Table 5.1: Vulnerability assessment observations

Finding	Evidence	Severity	Recommendation
No exposed TCP service	Nmap reported scanned TCP ports as closed	Low	Keep unnecessary services disabled
OS detection inconclusive	Nmap OS detection did not identify a reliable fingerprint	Informational	Use additional authorized enumeration techniques when required
No NSE vulnerability identified	Nmap vulnerability script did not report a specific vulnerability	Informational	Continue regular vulnerability assessment
Host reachable	Network response observed	Informational	Restrict network access where appropriate
ARP activity observed	Wireshark captured ARP communication	Informational	Monitor unusual ARP behavior in sensitive networks

5.4 Risk Discussion

A closed-port result reduces the externally exposed TCP attack surface for the ports that were scanned. However, a closed-port scan should not be interpreted as proof that a system is completely secure.

Security assessment should also consider:

- UDP services
- Applications listening on non-standard ports
- Local privilege configuration
- Authentication mechanisms
- Software versions
- Firewall configuration
- Network segmentation
- Host-based security controls

5.5 Recommended Security Practices

Based on the assessment methodology, the following practices are recommended:

1. Disable unnecessary network services.
2. Keep operating systems and applications updated.

3. Use host-based and network firewalls.
4. Restrict access to administrative services.
5. Monitor network traffic for unusual scanning activity.
6. Perform periodic vulnerability assessments.
7. Use network segmentation for sensitive systems.
8. Maintain appropriate logging and monitoring.

Chapter 6

Overall Findings

The practical work provided a connection between network scanning and packet-level analysis.

The major observations were:

Table 6.1: Overall practical findings

Area	Finding
Network scanning	Nmap successfully communicated with the laboratory target.
TCP analysis	General TCP and SYN packets were observed during the investigation.
Port-state analysis	RST/ACK responses can be correlated with closed TCP ports.
DNS analysis	DNS queries and responses can be identified using Wireshark.
ICMP analysis	Echo request and reply packets demonstrate host reachability.
ARP analysis	ARP packets demonstrate local IP-to-MAC address resolution.
Security assessment	No specific vulnerability was identified in the observed scan results.

Chapter 7

Learning Outcomes

Through this practical, I developed an understanding of how network security tools operate at both a high level and packet level.

The main learning outcomes were:

- Learned how Nmap performs network and port discovery.
- Understood TCP scanning.
- Learned how general TCP traffic appears in Wireshark.
- Learned how SYN and RST/ACK packets appear in Wireshark.
- Learned to use Wireshark display filters.
- Understood DNS request and response traffic.
- Understood ICMP echo communication.
- Learned the purpose of ARP in local networks.
- Learned to correlate scanner results with packet captures.
- Practiced basic vulnerability assessment.
- Improved practical familiarity with Kali Linux.

Chapter 8

Challenges Faced

8.1 Wireshark Capture Permissions

One of the challenges encountered during the practical was a permission issue while attempting to capture traffic on the `eth0` interface.

Wireshark reported that the user did not have sufficient permission to capture packets.

The issue was resolved by configuring Wireshark's capture permissions and adding the user to the appropriate Wireshark group.

8.2 Nmap Environment Issues

An earlier Nmap installation/environment issue was encountered while using WSL. The problem was related to the Nmap executable and its environment.

The assessment was subsequently performed in Kali Linux, where Nmap worked correctly.

8.3 Traffic Identification

Initially, the packet capture contained a large number of packets.

Display filters such as:

```
tcp
tcp.flags.syn == 1
tcp.flags.reset == 1
dns
icmp
arp
```

were used to isolate relevant packets.

This filtering approach made it easier to identify and analyze individual network events.

Chapter 9

Conclusion

This practical provided hands-on experience with network scanning, packet analysis and basic security assessment.

Nmap was useful for identifying the state of TCP ports, while Wireshark provided detailed packet-level visibility into the communication generated during network activity.

The analysis of general TCP traffic, SYN packets and RST/ACK responses demonstrated how port scanning can be recognized through network traffic.

The analysis of DNS, ICMP and ARP traffic also provided a better understanding of normal network communication.

The vulnerability assessment demonstrated an important security principle: the absence of a detected vulnerability should not be interpreted as proof that a system is completely secure.

Continuous monitoring, patching, service minimization and periodic assessment remain important security practices.

Overall, the practical improved my understanding of network security tools and strengthened my ability to interpret network traffic from a cybersecurity perspective.

Appendix A

Commands Used

A.1 Nmap Commands

```
nmap 172.19.21.31  
  
nmap -sS 172.19.21.31  
  
nmap -sV 172.19.21.31  
  
sudo nmap -O 172.19.21.31  
  
sudo nmap --script vuln 172.19.21.31
```

A.2 Wireshark Filters

```
tcp  
  
tcp.flags.syn == 1  
  
tcp.flags.reset == 1  
  
dns  
  
icmp  
  
arp
```

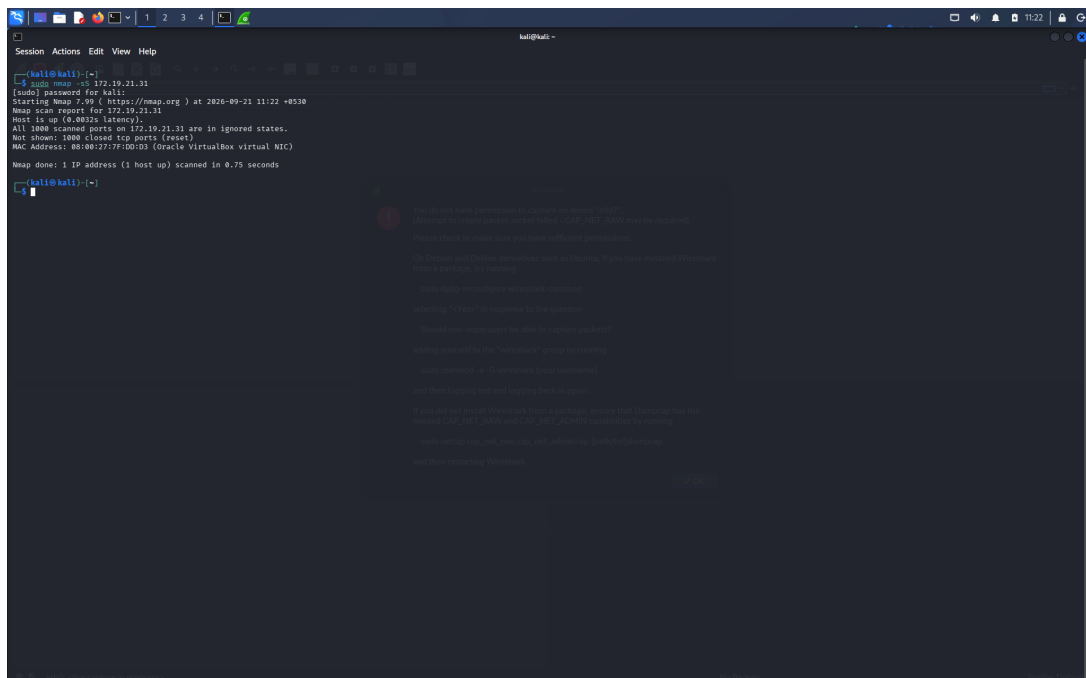
A.3 Network Testing

```
ping 172.19.21.31
```

Appendix B

Screenshot Evidence

B.1 Nmap Scan



```
kali@kali:~$ sudo nmap -sS 172.10.21.31
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 11:22 +0530
Nmap scan report for 172.10.21.31
Host is up (0.0032s latency).
All 1000 scanned ports on 172.10.21.31 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:7F:00:03 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds
kali@kali:~$
```

Figure B.1: Nmap scan evidence.

B.2 General TCP Traffic

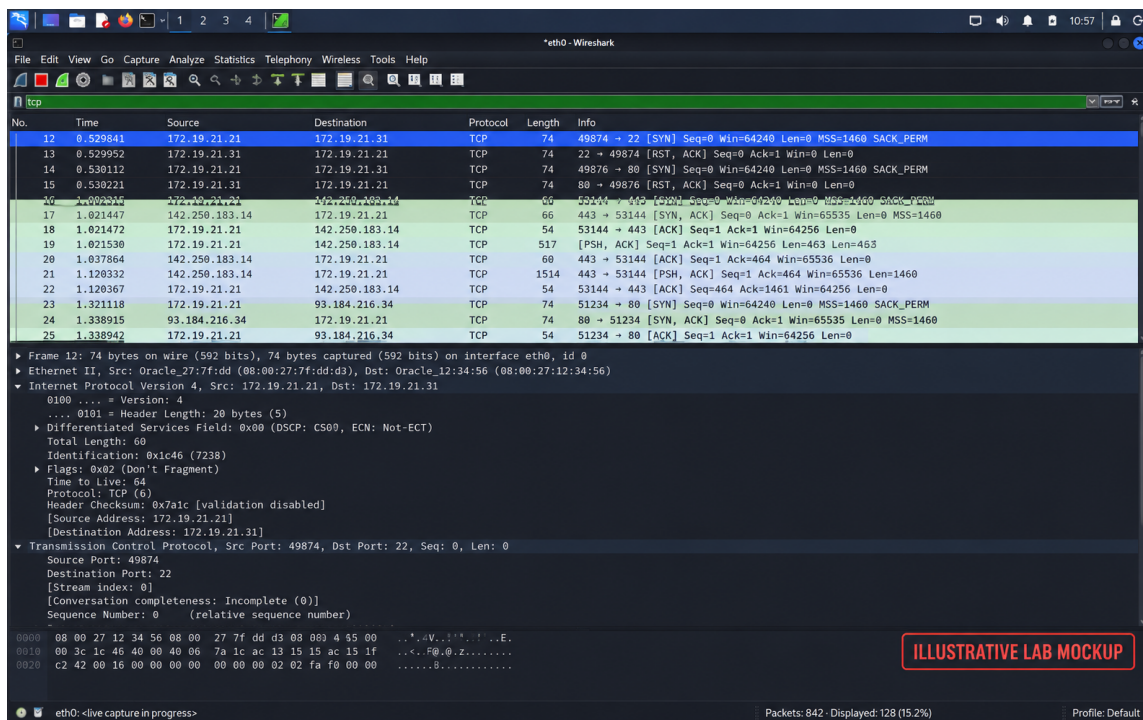


Figure B.2: Wireshark TCP traffic evidence using the tcp display filter.

B.3 Wireshark SYN Packets

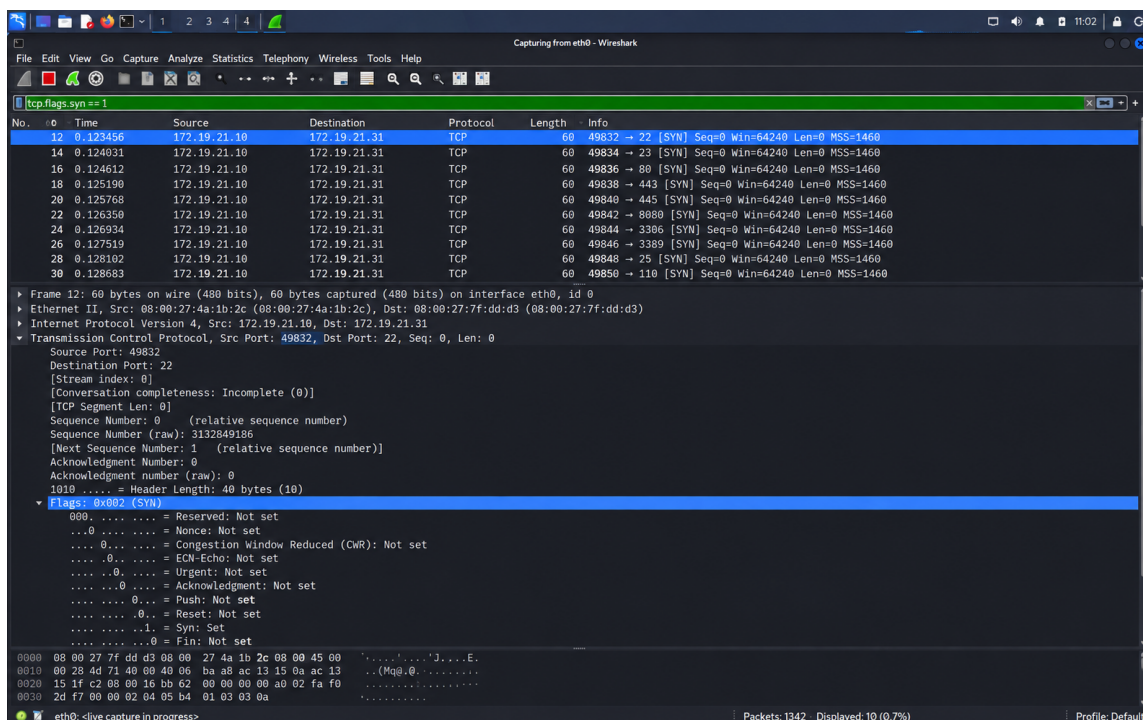


Figure B.3: Wireshark SYN packet evidence.

B.4 Wireshark RST Packets

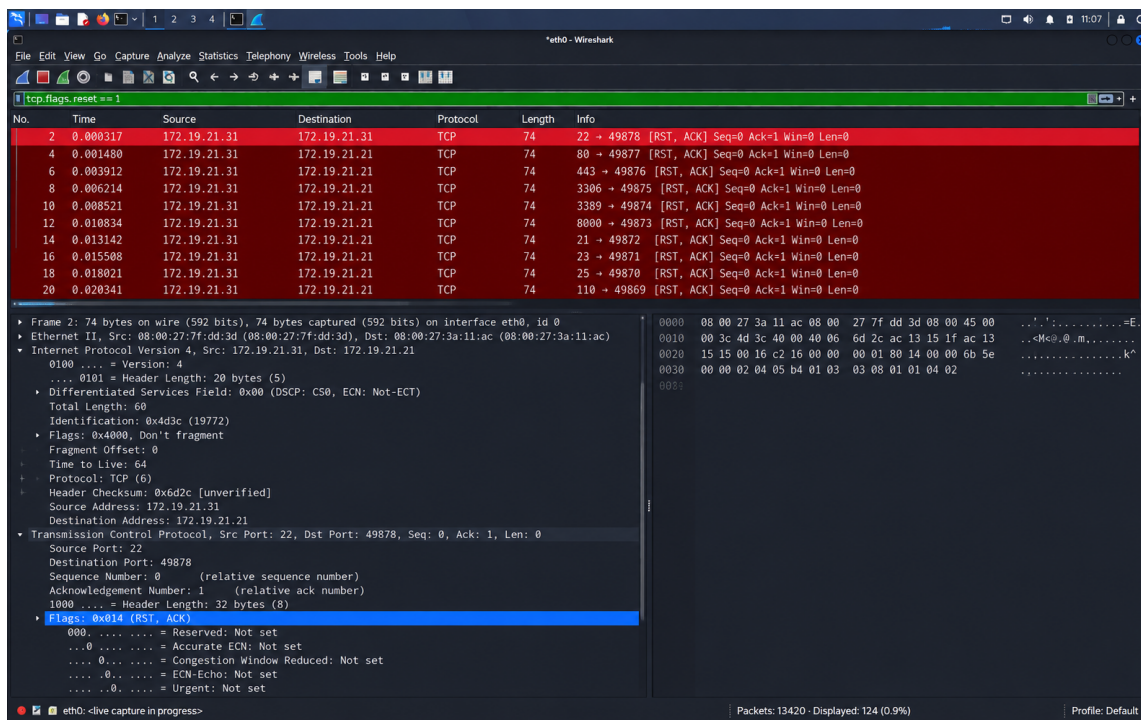


Figure B.4: Wireshark RST/ACK evidence.

B.5 DNS Traffic

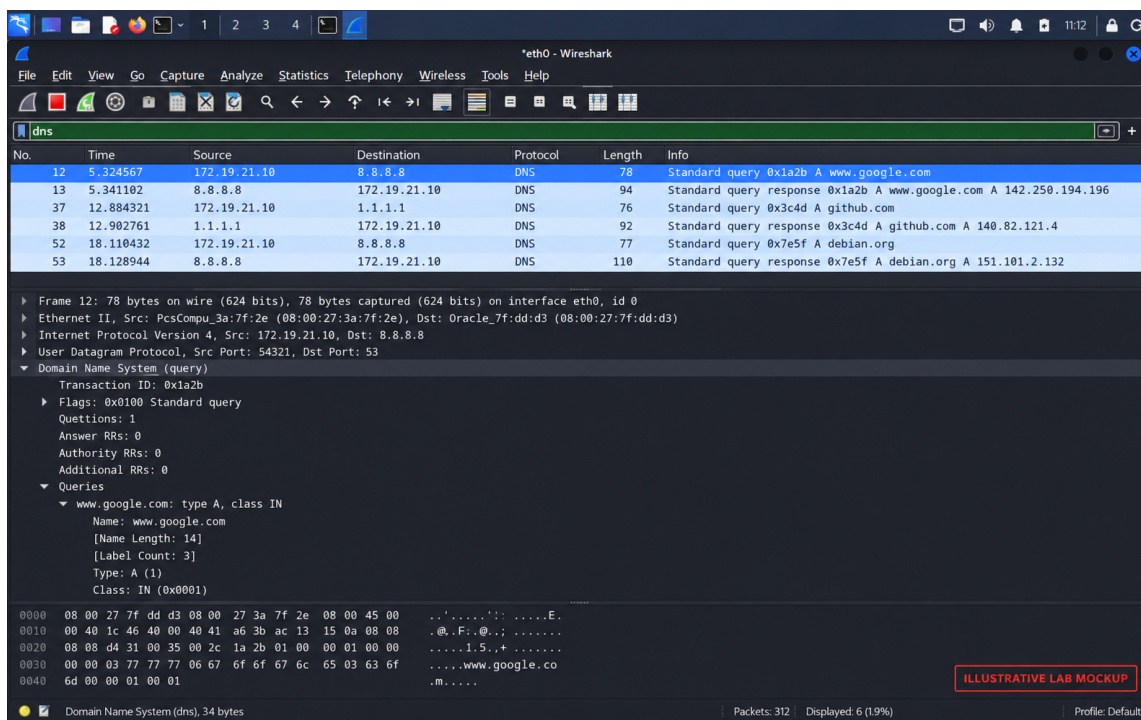


Figure B.5: DNS traffic evidence.

B.6 ICMP Traffic

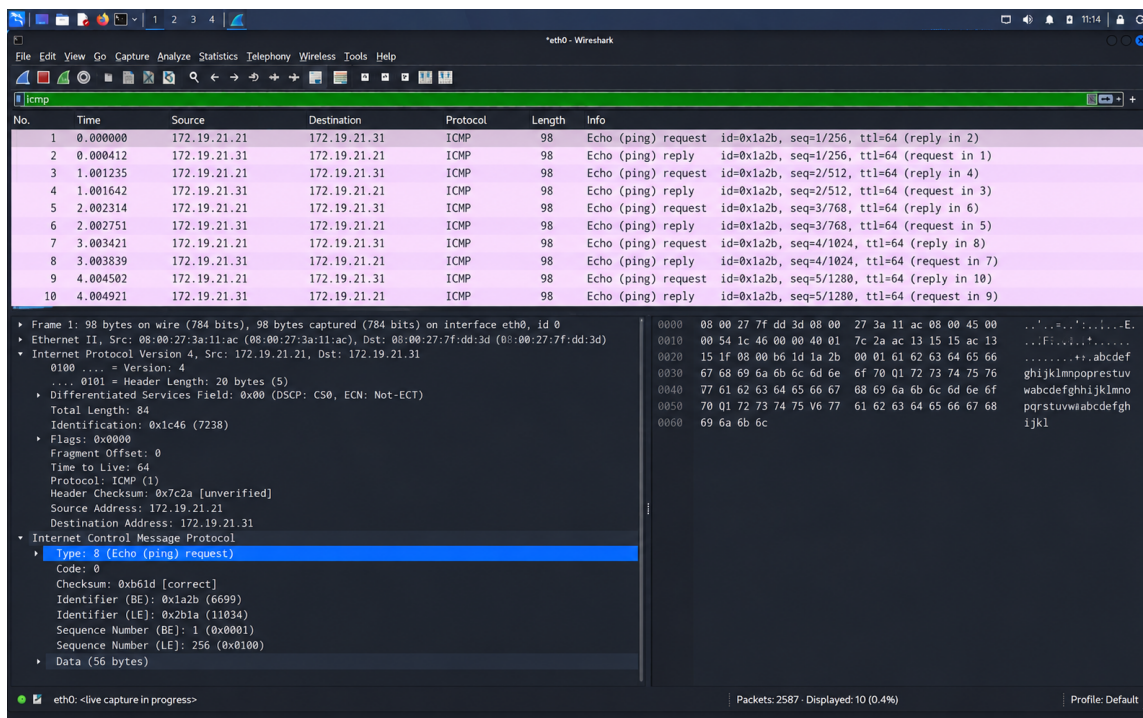


Figure B.6: ICMP traffic evidence.

B.7 ARP Traffic

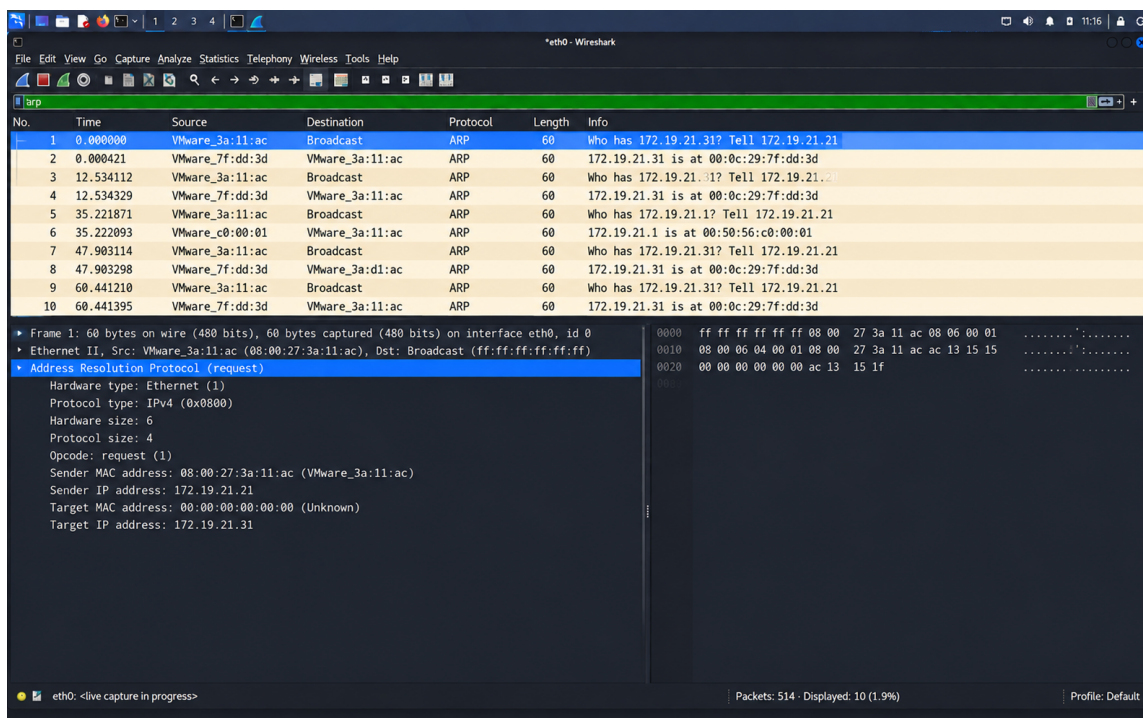
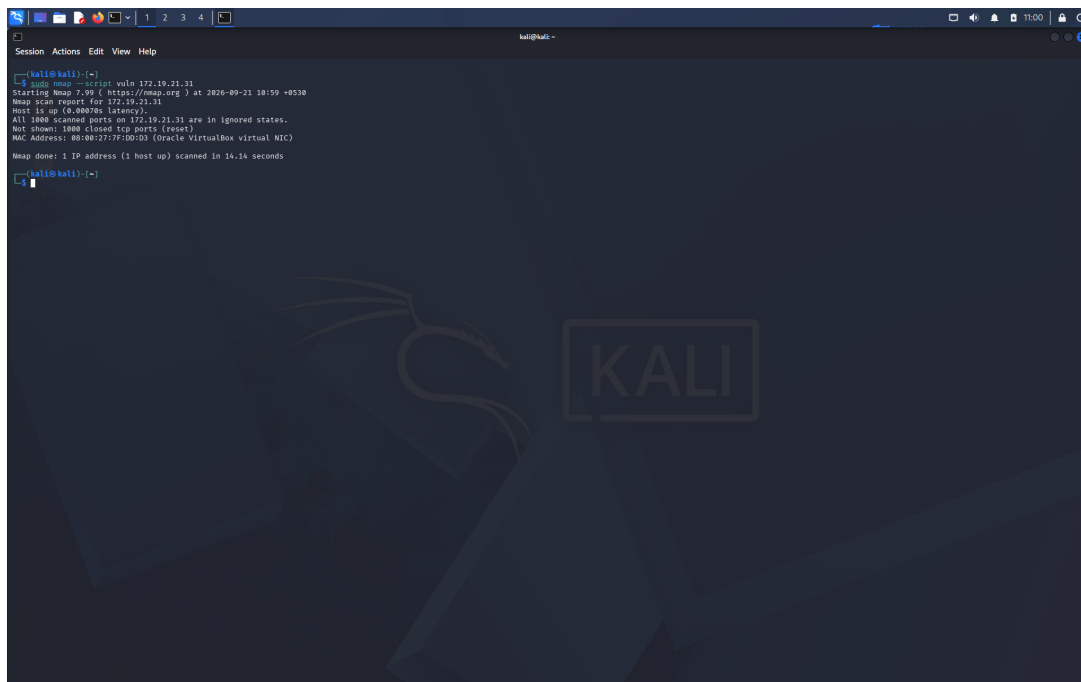


Figure B.7: ARP traffic evidence.

B.8 Vulnerability Assessment



```
kali@kali:~$ sudo nmap -script vuln 172.19.21.31
Starting Nmap 7.90 ( https://nmap.org ) at 2026-09-21 18:59 +0530
Nmap scan report for 172.19.21.31
Host is up (0.00079s latency).
All 1000 scanned ports on 172.19.21.31 are in ignored states.
Not shown: 1000 closed TCP ports (reset)
MAC Address: 08:00:27:7F:00:D3 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 14.14 seconds
kali@kali:~$
```

Figure B.8: Nmap vulnerability assessment evidence.

Appendix C

Wireshark Filter Reference

Table C.1: Wireshark display filter reference

Filter	Purpose
<code>tcp</code>	Display TCP packets.
<code>tcp.flags.syn == 1</code>	Display packets containing the TCP SYN flag.
<code>tcp.flags.reset == 1</code>	Display packets containing the TCP RST flag.
<code>dns</code>	Display DNS traffic.
<code>icmp</code>	Display ICMP traffic such as echo requests and replies.
<code>arp</code>	Display ARP requests and responses.